

AI Procurement Platform

3-Week Strategic Implementation Plan

Project Overview

This document outlines the accelerated development roadmap for an AI-powered Procurement Transformation platform. The plan integrates a modern tech stack with the 7-step procurement lifecycle (Identification to Payment) and advanced AI capabilities including Contract Analysis, Spend Analytics, and Supplier Intelligence.

Technology Stack

Next.js 15 FastAPI LangGraph Azure OpenAI (GPT-4o) MongoDB Atlas Azure Static Web Apps

WEEK 1: Foundation & AI Infrastructure

Goal: Establish core environment, RAG pipeline, and base application shell.

Day	Focus Areas & Key Deliverables
Day 1-2	Architecture & Core Models: - Set up FastAPI & Next.js 15 repositories. - Initialize MongoDB collections (Users, Clients, Suppliers, Contracts). - Configure Azure App Service, Blob Storage, and OpenAI deployments.
Day 3-4	Data Ingestion & RAG Base: - Implement PDF/CSV upload pipeline to Azure Blob. - Setup LangChain loaders and MongoDB Vector Search. - Deploy base RAG Chatbot for procurement document Q&A.
Day 5	Frontend Foundation: - Dashboard shell with shadcn/ui and Sidebar navigation. - Integration of Auth middleware and global state management.

WEEK 2: AI Agents & Procurement Workflows

Goal: Build intelligent automation for the end-to-end procurement lifecycle.

Day	Focus Areas & Key Deliverables
Day 6-7	Analysis Agents: • Contract Analysis: Clause extraction, risk detection, and compliance. • Spend Analytics: Anomaly detection and savings recommendations.
Day 8-9	Intelligence & Advisory: • Supplier Intelligence: ESG scoring, risk prediction, and performance tracking. • Procurement Advisor: Multi-agent LangGraph workflow for policy guidance.
Day 10	Workflow Engine: • Implementation of the 7-step flow: Requisition → Sourcing → PO → Goods Receipt → Invoice → Payment.

WEEK 3: Enterprise Features & Production Readiness

Goal: Finalize multi-tenancy, security, and executive reporting.

Day	Focus Areas & Key Deliverables
Day 11-12	Multi-Tenancy & Auth: • Client Management module with tenant isolation. • RBAC implementation (Admin, Procurement, Finance, Auditor).
Day 13	Executive Dashboard: • KPI widgets for Spend Trends, Supplier Heatmaps, and Cycle Metrics. • AI-generated summary reports and CSV/PDF exports.
Day 14-15	Optimization & Handover: • QA, performance tuning, and Azure Monitoring setup. • Final UAT, documentation, and production deployment.

Measurable Impact Targets

- **Sourcing Efficiency:** 60% reduction in cycle time via AI-driven RFQ.
- **Accuracy:** 40% fewer invoice errors with ML 3-way matching.

- **Savings:** 3-8% additional cost identification through AI analytics.
- **Compliance:** 80% faster contract review using NLP intelligence.